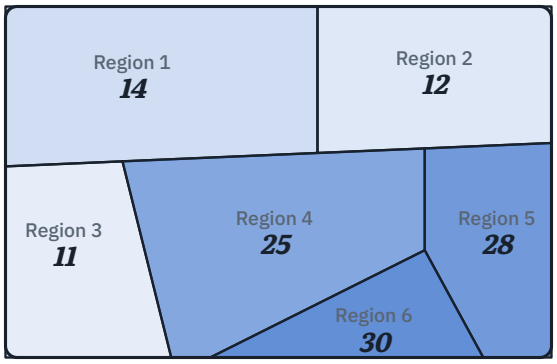
Moran's *I* by Hand

A worked example for six regions. Each step shows the full Moran's I formula with the part being calculated highlighted in red, a short explanation, and the arithmetic.

The problem

MAP AND VALUES x_i



Six regions; darker shading means a higher value.

BINARY CONTIGUITY WEIGHTS w_{ij}

	1	2	3	4	5	6	Row sum
1	0	**1**	**1**	**1**	0	0	**3**
2	**1**	0	0	**1**	**1**	0	**3**
3	**1**	0	0	**1**	0	0	**2**
4	**1**	**1**	**1**	0	**1**	**1**	**5**
5	0	**1**	0	**1**	0	**1**	**3**
6	0	0	0	**1**	**1**	0	**2**
					TOTAL		**18**

$w_{ij} = 1$ if regions *i* and *j* share a border, otherwise $w_{ij} = 0$. A region is not its own neighbor.

THE FORMULA

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Steps

STEP 1 Count the regions

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}\right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Moran's I scales the result by the number of regions, n. Count the areal units in the study region.

WORK

$n = 6$

RESULT $n = 6$

STEP 2 Compute the mean

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}\right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Every comparison in Moran's I is made against the average value across all regions. Add the six values and divide by n.

WORK

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i = \frac{14 + 12 + 11 + 25 + 28 + 30}{6} = \frac{120}{6}$$

RESULT $\bar{x} = 20$

STEP 3

Find each region's deviation from the mean

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Subtract the mean from each value. A negative deviation marks a below-average region and a positive deviation an above-average region. These deviations appear in both the numerator and the denominator.

WORK

Region i				
1	14	−6		36
2	12	−8		64
3	11	−9		81
4	25	5		25
5	28	8		64
6	30	10		100
			TOTAL	370

RESULT $x_i - \bar{x} = \{-6, -8, -9, 5, 8, 10\}$

STEP 4 Sum the spatial weights

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Add every entry of the weight matrix. With binary weights this counts the neighbor links, and each pair of neighbors is counted twice (once as *i*–*j* and once as *j*–*i*). The row sums give each region's number of neighbors.

WORK

$$\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} = 3 + 3 + 2 + 5 + 3 + 2$$

RESULT $\sum_i \sum_j w_{ij} = 18$

STEP 5 Sum the squared deviations

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Square each deviation and add them up. This measures the total variation in the data and standardizes the statistic, much like the denominator of a correlation coefficient.

WORK

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 = (-6)^2 + (-8)^2 + (-9)^2 + (5)^2 + (8)^2 + (10)^2$$

$$= 36 + 64 + 81 + 25 + 64 + 100$$

RESULT $\sum_i (x_i - \bar{x})^2 = 370$

STEP 6 Sum the cross-products of neighboring deviations

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}\right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

For every pair of neighbors ($w_{ij} = 1$), multiply their deviations. Same-sign pairs (both above or both below the mean) add to the sum; opposite-sign pairs subtract from it. Working row by row, add the deviations of region *i*'s neighbors, then multiply by region *i*'s own deviation.

WORK

Region i	$x_i - \bar{x}$	Neighbors j	$\sum_j w_{ij}(x_j - \bar{x})$	$(x_i - \bar{x}) \sum_j w_{ij}(x_j - \bar{x})$
1	−6	2, 3, 4	(−8) + (−9) + (5) = −12	−6 × -12 = **72**
2	−8	1, 4, 5	(−6) + (5) + (8) = 7	−8 × 7 = **−56**
3	−9	1, 4	(−6) + (5) = −1	−9 × -1 = **9**
4	5	1, 2, 3, 5, 6	(−6) + (−8) + (−9) + (8) + (10) = −5	5 × -5 = **−25**
5	8	2, 4, 6	(−8) + (5) + (10) = 7	8 × 7 = **56**
6	10	4, 5	(5) + (8) = 13	10 × 13 = **130**
			TOTAL	**186**

RESULT $\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x}) = 186$

STEP 7 Assemble Moran's I

$$I = \frac{n \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}\right) \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Substitute the pieces from Steps 1–6 into the formula.

WORK

$$I = \frac{6 \cdot 186}{18 \cdot 370} = \frac{1116}{6660}$$

RESULT $I = 0.1676$

Interpret the result

I = 0.1676 is positive, so regions tend to sit next to regions with similar values: the low values (Regions 1, 2 and 3) run along the north and west, and the high values (Regions 4, 5 and 6) are grouped in the south and east. The value is well below 1, so the positive spatial autocorrelation is present but not strong.

With no spatial autocorrelation we would expect $E[I] = \frac{-1}{n-1} = -0.2$, so the observed value sits above what a random arrangement would produce on average. Whether that difference is statistically significant requires the variance of I and a z-test.

Source: worked example adapted from P. A. Rogerson and I. Yamada (2008), *Statistical Detection and Surveillance of Geographic Clusters*, §3.4.1.